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Article information

Article title

Auditing source correspondence and observation coverage in robot IMU snapshot logs

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Keywords

robot measurement; inertial measurement unit; trace verification; latest snapshot; observation coverage; reproducibility

Related research article

[Authors to confirm related research article status before submission.]

Abstract

Robot applications often read a latest-value state repeatedly, complicating comparisons between retained source packets and application observations. We describe an offline method that verifies source checksums and sequence order, checks every application payload, and evaluates all order-preserving content correspondences. The procedure reports unique, unmatched or ambiguous correspondence and separates internal observation omissions from capture-edge frames. Four historical paired captures contain 4,378 source frames and 20,835 application reads, representing 4,326 distinct snapshots, 46 internal omissions and six edge frames. A unique-path recurrence agreed with independent enumeration in 7,340 short-sequence cases. All 224 controlled transformations and invariance controls produced their specified outcomes. These offline cases validate implementation behavior within a declared input contract; they are not additional robot experiments. The method supports trace qualification before downstream joint–inertial comparisons. It does not establish physical sensor accuracy, internal delivery causality or dynamic repeatability.

- Preserve every read while separating repeated IMU snapshots from advancing joint feedback.
- Withhold exact source attribution when multiple complete monotone correspondences remain.
- Reproduce observation accounting and controlled validation without operating a robot.

Graphical abstract

Provided as the separate graphical_abstract.pdf file.

Specifications table

Subject area	Engineering
More specific subject area	Robot IMU source and snapshot auditing
Name of your method	Uniqueness-aware source correspondence and observation accounting
Name and reference of original method	No single parent algorithm. Validation basis: S. Segura et al., A Survey on Metamorphic Testing, IEEE Transactions on Software Engineering 42 (2016) 805–824. DOI: 10.1109/TSE.2016.2532875.
Resource availability	Selected data, code and result tables accompany the private reviewer archive. [Authors to insert persistent repository accession and redistribution terms.]

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Background

Robot measurements pass through acquisition, parsing, state publication and application reading. These stages expose different observations. Response-time analysis models execution within processing chains [1], while runtime tracing provides execution events for investigating software behavior [2]. Message-flow analysis connects instrumented publications and callbacks, including complex causal relationships [3]. Such instrumentation is valuable when the relevant events were recorded.

Historical robot logs may instead retain source bytes, decoded state values and application timestamps without end-to-end identifiers. Data-refreshing semantics can overwrite earlier values before a reader observes them, as recognized in processing-chain timing analysis [4]. A recent survey distinguishes response time, reaction time and data age [5]. Accordingly, a small host timestamp difference cannot establish complete observation of retained source input.

Our practical problem is narrower than reconstructing a distributed execution. Before comparing an onboard joint signal with an inertial measurement unit (IMU), we need to know which retained source values appear in the application. Repeated reads must remain distinguishable from new observations. Repeated source values may also leave several order-preserving explanations. Selecting the earliest compatible value conceals that ambiguity.

We combine packet checks, all-row payload constraints, uniqueness-aware content matching and exact frame accounting into a reproducible offline procedure. The procedure uses established packet checks, dynamic programming and set accounting. Its contribution is an explicit input contract, typed outputs and validation for legacy robot records without shared acquisition timestamps. Controlled transformations test required output relationships, following the rationale of metamorphic testing [6]. The resulting report identifies which observations can support a downstream comparison and which must remain unresolved.

Method details

1 Define the input contract

Require a source stream with framed payloads, declared component units and a sequence field, together with an application CSV containing host time, IMU state time and decoded components. Preserve immutable originals and record SHA-256 hashes before analysis. Verify the logger and decoder definitions rather than pairing files by name, modification time or nearest wall-clock time.

The reference adapter handles YIS106 packets beginning with hexadecimal 59 53. Its two-byte sequence is little-endian and wraps modulo 65,536. A one-byte payload length precedes type–length–value components. Two cumulative 8-bit sums validate the sequence, length and payload bytes. This is an additive packet checksum, not a cyclic redundancy check. Unsupported component layouts raise a decoding error.

Ten comparison components comprise three accelerations, three angular rates and a scalar-first quaternion. Acceleration and quaternion integers are scaled by 10^{-6}. Angular-rate integers are scaled by 10^{-6} degrees per second and converted to radians per second. Euler-angle fields are decoded structurally but excluded from matching because the quaternion already represents orientation. The method assumes a new state timestamp represents a later retained source frame. Resampled, interpolated or many-to-one state updates require a different adapter contract.

2 Check source and application observations separately

Decode complete packets and report checksum failures, sequence anomalies, leading discarded bytes and incomplete trailing bytes. A source-integrity result requires at least one complete packet. Consecutive sequence values alone cannot establish that every physical measurement was generated or captured. Capture-edge bytes remain outside complete-frame counts.

Check every application row for required fields, finite components and positive IMU timestamps. Require nondecreasing application host time and IMU state time, allowing repeated state timestamps. For each state timestamp, require every component range across its rows to remain within the declared numerical tolerance. A conflicting tuple rejects that application record before grouping. An old timestamp recurring after a newer one also fails clock ordering.

Group only the IMU tuple by its timestamp after these checks. Keep the original joint rows for interpolation or subsequent analysis. The primary absolute tolerance is 10^{-10} in each stored component unit. It covers numerical serialization differences and is not a physical sensor-error tolerance. Recheck conclusions at 10^{-12}, 10^{-8} and 10^{-6} rather than selecting a tolerance to obtain a preferred match.

3 Resolve complete source correspondence

Number complete source frames from zero. For observed timestamp group i, form candidate set C_i from source indices whose ten components differ by at most the tolerance from every row in that group. Componentwise group minima and maxima implement this all-row constraint. Testing only the first row can overlook a later discrepancy near the tolerance boundary.

A complete correspondence chooses one candidate per group with strictly increasing source indices. Count complete paths using the following recurrence, with the first layer initialized to one at each candidate. Counts are capped at two because only zero, one or multiple explanations are needed.

$$c_i(j) = \min\left(2, \sum_{k \in C_{i-1},\ k < j} c_{i-1}(k)\right)$$

The final capped sum is zero for unmatched, one for unique and two for ambiguous correspondence. Reconstruct source indices only for the unique case. Local candidate multiplicity does not imply global ambiguity because later observations may eliminate an earlier choice. Conversely, an earliest-forward join can return one path while another complete path remains (Figure 1).

The implementation uses a componentwise distance query to generate candidates, then verifies group extrema explicitly. Sorted predecessor counts are accumulated with a running prefix sum. Backpointers reconstruct a unique path without enumerating every complete path. The independent validation oracle instead enumerates source-index combinations directly. Candidate density governs resource use; performance on these short records does not establish a deployment-time bound.

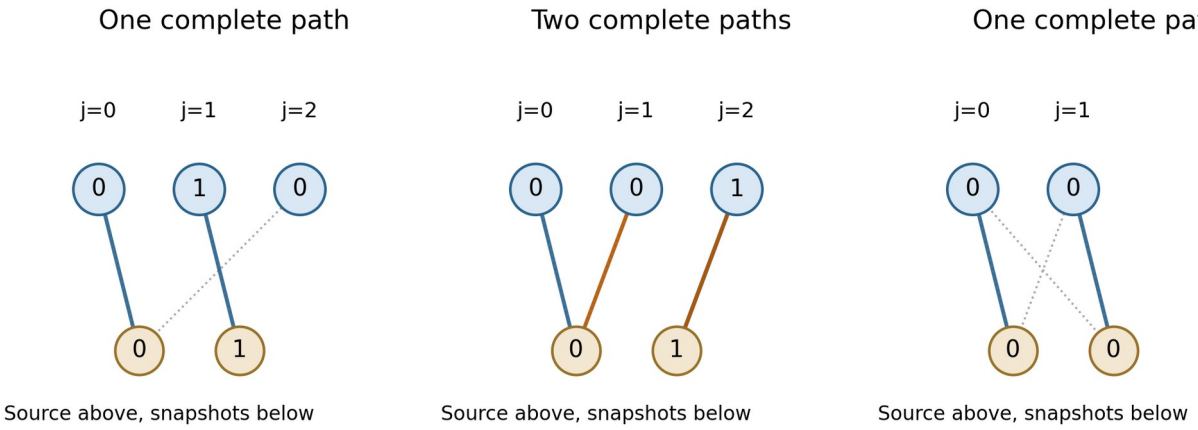


Figure 1. Small illustrative matching graphs. Values are synthetic symbols. Dotted lines indicate compatible values; colored lines belong to complete strictly increasing correspondences. Left: a later observation resolves a local alternative. Middle: two complete paths remain. Right: ordering resolves identical values when both source frames must be represented. These examples are logical fixtures, not robot recordings.

4 Account for retained but unobserved source frames

For a unique path, let F be complete source frames and N distinct observed IMU timestamps. Let S sum the source-index gaps between adjacent matches, excluding their endpoints. Let E count frames before the first match and after the last match. Every retained frame then belongs to exactly one of three sets.

$$F = N + S + E$$

Report S as internal unobserved source frames, not UART loss or driver rejection. The raw capture and application state need not reveal what occurred between those observation points. Do not report exact attribution when the match is ambiguous. Sequence anomalies or checksum errors invalidate the intact-source accounting prerequisite.

5 Interpret clocks and qualify downstream comparisons

In the inspected logger, application host time is recorded immediately before reading IMU and joint state. Inspected driver source assigns the IMU timestamp on the robot after parsing. Their difference is a host receipt-age diagnostic. Hardware timestamp fields are zero, and the source bridge clock is unsynchronized with the robot host. Never subtract those cross-machine clock values or interpret the host difference as measurement age at the sensor.

Unique correspondence qualifies retained content for a comparison under the stated contract. It does not certify deployed binary identity, mounting geometry or causal passage through every internal stage. Any joint comparison must separately verify reference coordinates, available components and excitation. A coordinated motion or another sensor installation is not automatically a repetition of an isolated-joint protocol.

6 Reproduce the audit offline

The accompanying reviewer archive includes the six original application CSVs, five source captures, chunk records, adapter, case generator and result tables. Run python code/reproduce.py from the extracted archive using the recorded dependencies. The command recomputes matching, controlled validation, the downstream illustration and

figures without network or hardware access. A single-pair command and output schema are provided in the supplement. Frozen inputs and the expected result hashes identify the analyzed version.

Method validation

Original record correspondence and coverage

The retrospective corpus contains three setup/static pairs, one paired waist-J3 excursion and return, one source-only capture, and two software-only records. All five source captures were compared with all six software records. Exactly four of 30 combinations had unique complete correspondence at every tested tolerance. The other 26 were unmatched. The source-only capture contained 1,000 complete frames, four leading bytes and 32 incomplete trailing bytes; it supplies no application-coverage evidence.

Table 1. Accounting of the four original paired records. Reads include repeated IMU snapshots; all source checksums and consecutive sequence checks passed. N, S and E partition F.

Record	F	Reads	N	S	E	Max receipt age ms
S1	802	3,883	800	0	2	10.114
S2	802	3,835	796	5	1	10.106
S3	802	3,821	795	5	2	10.077
D1	1972	9,296	1,935	36	1	10.077

Across the admitted pairs, 16,509 repeated reads account for the difference between application rows and distinct IMU timestamps. All reads satisfy their source-component constraints, with maximum absolute discrepancy 1.78×10^{-15}. Their complete-frame identity is 4,378 = 4,326 + 46 + 6. Counts describe these retained traces rather than independent robot trials.

The motion record contained a 320.010 ms interval spanning 31 unobserved source frames, while the maximum host receipt age was 10.077 ms. Another interval contained five unobserved frames. Thus, application coverage and the host receipt-age diagnostic give different information. No source-index omission occurred inside either logged motion-phase interior. The source sequences were continuous throughout all four retained pairs (Figure 2).

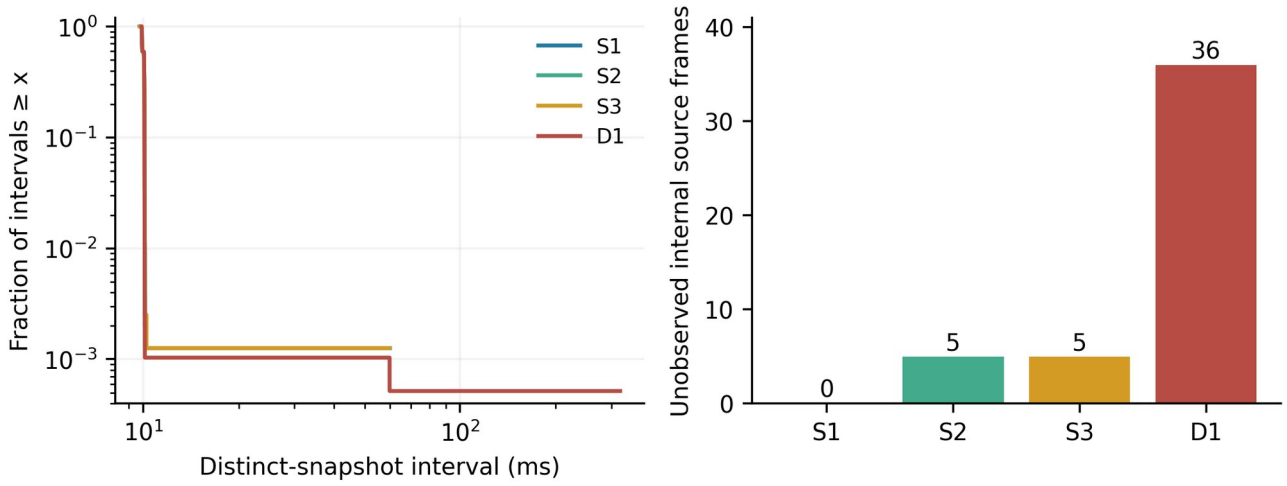


Figure 2. Coverage in the four original paired records. Left: empirical tail fractions of distinct-snapshot intervals on logarithmic axes. Right: source-index counts excluded from application observation within capture interiors. These finite traces differ in setup and acquisition stage; the plot does not estimate a causal configuration effect or physical packet-loss rate.

Independent enumeration and controlled transformations

The path recurrence agreed with independent enumeration in all 7,340 tested binary-sequence cases. Source lengths ranged from one to seven, and observation lengths from one to four, bounded by source length. The oracle checked both the capped path count and reconstructed indices. This is a finite combinatorial implementation check, not a proof that every possible input is handled correctly.

For each original pair, controlled local copies used interior anchors at 25%, 50% and 75% of observed groups. Omission and duplicate-read sizes were 1, 5, 20 and 50. Other transformations changed payloads, clocks or source packets. The validation matrix and development-driven additions are documented in the supplement. Original records were never overwritten.

All 224 cases produced their specified typed outcomes. This total includes 56 unaltered or benign invariance controls and 168 other transformations. Removing k internal observed groups increased S by exactly k in all 48 omission cases. Adding repeated reads preserved correspondence and S in all 48 duplicate-read controls. Valid sequence-wrap controls remained admissible.

Table 2. Representative controlled families. Last three columns count passes of deliberately limited checks, not correct full-method decisions. The full method met every specified outcome. Duplicate reads and valid sequence wraps are invariance controls; omissions require quantitative coverage reporting.

Controlled family	Cases	Receipt age pass	Source packet pass	Earliest join pass
Omitted snapshots	48	48	48	48
Duplicate reads	48	48	48	48
Conflicting repeated tuple	12	12	12	12
Tolerance chaining	12	12	12	12

Controlled family	Cases	Receipt age pass	Source packet pass	Earliest join pass
Old timestamp recurrence	12	0	12	12
Aliased source packet	12	12	12	12
Corrupted checksum	12	12	0	0
Valid sequence wrap	4	4	4	4

Three deliberately limited routines illustrate complementary check coverage: receipt age within 0–20 ms, valid source packets with consecutive sequence, and earliest-forward joining after first-row deduplication. All three checks passed all 48 omission cases because those edits preserve fresh surviving values and intact source input. The full procedure quantified the resulting coverage change.

Earliest-forward joining also returned a path in all 12 aliased-source cases, although multiple complete correspondences remained. The source insertion deliberately regenerated sequence numbers and checksums. Every-repeat payload constraints rejected 12 conflicting-repeat cases and prevented 12 tolerance-chain cases from receiving a complete match. These designed counts are diagnostic examples, not estimates of sensitivity, specificity or field reliability.

Illustrative joint and inertial comparison

The single J3 excursion illustrates downstream use after correspondence qualification. Supplementary methods retain sign, gain and shift controls with 169 distinct observations in each direction. Those directions belong to one acquisition, and the resulting channel disagreement does not establish accuracy, identified hardware latency or dynamic repeatability.

Limitations

Content correspondence requires the declared decoding units, a compatible component layout and monotone state-update semantics. Unknown transformations, missing state timestamps or resampling may prevent matching. Repeated source values can remain ambiguous even with intact records; the method must then withhold exact attribution. Unique value matching is conditional evidence, not proof of internal causal delivery.

The real paired corpus comes from one robot and a limited set of historical configurations. Its independent dynamic evidence is one single-axis excursion and return. Controlled edits and exhaustive short sequences strengthen software validation but do not expand physical replication or establish typical failure rates. The validation suite was developed with knowledge of the original traces and includes declared development-driven additions.

An absolute tolerance mixes components expressed in different units and is appropriate here for deterministic serialization correspondence only. Physical error assessment needs component-specific uncertainty models and external references. Unsynchronized source clocks and absent hardware acquisition timestamps preclude physical end-to-end latency estimates. Deployment identity and historical mounting information remain incompletely documented.

The implementation has not been evaluated prospectively on new logger formats, independent robots or continuous deployment. A useful next validation would freeze the adapter and criteria before analyzing a separately acquired corpus. That study should include explicit source identifiers or instrumented causal events alongside repeated, independently referenced motion sessions.

Ethics statements

This retrospective analysis used robot sensor and software records. No human participants, animal experiments or social-media data were studied. This revision performed local analysis only and did not operate a robot.

CRediT author statement

[Verified author contributions to be completed.]

Acknowledgments

[Funding and acknowledgments to be completed by the authors.]

Declaration of interests

[Author completion required: select one statement below and supply details if applicable.]

☐ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

[Authors to complete or mark not applicable after confirming interests.]

Supplementary material *and/or* additional information [OPTIONAL]

Supplementary methods provide input definitions, proof details, the full controlled-transformation table, tolerance results and the bounded downstream illustration. The private reviewer reproducibility archive contains selected original data and executable analysis. Repository accession, reviewer access and redistribution terms require author completion before submission.

References

[1] D. Casini, T. Blaß, I. Lütkebohle, B.B. Brandenburg, Response-Time Analysis of ROS 2 Processing Chains Under Reservation-Based Scheduling, 31st Euromicro Conference on Real-Time Systems 133 (2019) 6:1–6:23. https://doi.org/10.4230/LIPIcs.ECRTS.2019.6

[2] C. Bédard, I. Lütkebohle, M. Dagenais, ros2_tracing: Multipurpose Low-Overhead Framework for Real-Time Tracing of ROS 2, IEEE Robotics and Automation Letters 7 (2022) 6511–6518. https://doi.org/10.1109/LRA.2022.3174346

[3] C. Bédard, P.-Y. Lajoie, G. Beltrame, M. Dagenais, Message flow analysis with complex causal links for distributed ROS 2 systems, Robotics and Autonomous Systems 161 (2023) 104361. https://doi.org/10.1016/j.robot.2022.104361

[4] Y. Tang, X. Jiang, N. Guan, X. Luo, M. Yang, W. Yi, Timing analysis of processing chains with data refreshing in ROS 2, Journal of Systems Architecture 155 (2024) 103259. https://doi.org/10.1016/j.sysarc.2024.103259

[5] D. Casini, J.-J. Chen, J. Li, F. Reghenzani, H. Teper, A Survey of Real-Time Support, Analysis, and Advancements in ROS 2, Leibniz Transactions on Embedded Systems 11 (2026) 1:1–1:37. https://doi.org/10.4230/LITES.11.1.1

[6] S. Segura, G. Fraser, A.B. Sánchez, A. Ruiz-Cortés, A Survey on Metamorphic Testing, IEEE Transactions on Software Engineering 42 (2016) 805–824. https://doi.org/10.1109/TSE.2016.2532875